

```
/* Postfix expression evaluation */
```

```
#include<stdio.h>
```

```
#include <ctype.h>
```

```
int stack[20];
```

```
int top = -1;
```

```
void push(int x)
```

```
{
```

```
    stack[++top] = x;
```

```
}
```

```
int pop()
```

```
{
```

```
    return stack[top--];
```

```
}
```

```
int main()
```

```
{
```

```
    char exp[20];
```

```
    char *e;
```

```
    int n1,n2,n3,num;
```

```
    printf("Enter the expression :: ");
```

```
    scanf("%s",exp);
```

```
    e = exp;
```

```
    while(*e != '\0')
```

```
{
    if(isdigit(*e))
    {
        num = *e - 48;
        push(num);
    }
    else
    {
        n1 = pop();
        n2 = pop();
        switch(*e)
        {
            case '+':
            {
                n3 = n1 + n2;
                break;
            }
            case '-':
            {
                n3 = n2 - n1;
                break;
            }
            case '*':
            {
                n3 = n1 * n2;
```

```
        break;
    }
    case '/':
    {
        n3 = n2 / n1;
        break;
    }
}
push(n3);
}
e++;
}
printf("\nThe result of expression %s = %d\n\n",exp,pop());
return 0;
}
```